

Active Inference ~ Parr, Pezzulo, Friston ~ Chapter 7 ~ BookStream #002.0

BookStream | Parr, Pezzulo, Friston ~ Chapter 7 ~ BookStream #002.02 | 22:01

Chapter 7 is called Active Inference and Discrete Time.

Chapter 7 is the first in a pair of chapters with Chapter 8 on discrete and continuous time.

So they're kind of like two forks of a river that we discussed in Chapter 4 and before.

And described the recipe in Chapter 6.

Now 7 and 8 are kind of like one level deeper.

Going from the kind of all of this group of animals to one level deeper into its classification scheme.

On the way to the specific generative model for which it's actually given in its totality.

But everything prior to that is about the learning about its principles.

And this is kind of on the trunk of the path to discrete time modeling.

Just like Chapter 8 will be about continuous time modeling.

What would you add in?

Okay, so I think Chapters 7 and 8 really helps to understand in a more practical way.

how the materials from particularly Chapters 1 through 5 applies in real-time situations.

So even if we somehow didn't get to understand every detail of Chapters 1 through 4.

When we come to Chapters 7 and 8, I think some of those uncertainties about our understandings can be clarified,

at least in a practical sense.

So I believe these two chapters are really helpful in order to consolidate our understandings from the previous chapters.

Awesome. Well said.

So it's going to involve specifying some discrete time models.

7.2 goes into perceptual processing.

And the general structure of the chapter is going to walk through a series of examples that build in complexity,

where they first start with perception in 7.2,

introduce decision-making,

and then describe a few more types of motifs or cognitive structure or patterns.

And also check out step-by-step and model stream 1,

where it's built up to in a different way.

So the first example is,

I'll let you describe it,

since it's musical.

Okay.

So, yeah,

the first example is the situation in which we try to describe the performance of an amateur musician

in terms of how we listen to the performance of an amateur musician

in terms of the predictions we get from our anticipation of the following notes,

as opposed to the actual notes that's being played.

So these kinds of anticipatory reaction,

listening reaction to the musician can be successfully formalized using discrete time active inference

by putting together the matrices A for the states

and matrix B for the transition between the states

or the transition probabilities,

which in this case describes the probability from going from one node to the other.

And obviously the actual sequence that's been played,

which can be described with the matrix D .

So,

and another point I wanted to point,

I wanted to mention is,

for anyone who has downloaded this chapter before,

I don't know,

I think about June or something,

I recommend redownloading it from MIT's website

because they have corrected some of the typos

that was previously present in this chapter,

particularly in figure 7.2.

Cool.

So,

this graphical model

where a person is listening,

this is a general perceptual Bayesian framing,

it's specified.

Just like with any other equations,

there's a lot to look into,

but

A indicates the probability of an outcome

given a state.

This is saying,

if it were all

on the diagonal,

like an identity matrix,

this is kind of a common motif,

then states kind of map to themselves.

So,
in the context of,
in the context of this model,
A represents the mapping between the observed note
and the underlying hidden true note,
and then B describes the transition matrix
of how those change to time,
D is the prior.

They're specified.

Figure 7.2,
do you want to describe it?

All right.

So,
in figure 7.2,
or at least
the incomplete version
of figure 7.2
we see here,
well,

at the upper left
part of the picture,
we see,

I mean,
the beliefs about,
about each note
at each step,
at each time step,
and at upper right,
we somehow translate
those beliefs
into specific numerical values.

So,
instead of just
assigning some continuous values,
we've simplified
the situation
by assigning some
discrete numerical values

for each note.
and then
the lower left
is supposed to show
the free energy gradients
over time,
or in other terms,
the prediction errors
we get
from,
I mean,
comparing our predictions
with the actual outcomes.
So,
lastly,
the lower right picture
shows,
in parallel
to the upper right picture,
determines
the values
of these errors.
so,
we can see both
the continuous,
the initial,
or at least
initial,
continuous
assignment
and values,
and then
the further
discretizing
of the values
in order to
get the
discrete time

situation,
or the more
tractable
discrete time
situations.

Okay.

So,
it's a general
passive
inference
task
where
there's
priors
about
how
states
are
going
to
change
through
time,
and then
there's
real data
coming in.

So,
that's
the kind
of
classical
predictive
coding,
video
compression,
Kalman
filter,

Bayesian

setting.

7.3

introduces

a key

motif,

which is

decision-making

and planning

as

inference.

So,

this is

the idea

of having

a Bayes

graph

where the

variables can

relate to

different things.

There's

high

composability.

And here,

the idea

is that

a variable

is going

to be

proposed

that we

can do

inference

about

that

describes

the process

of decision-making

or policy

selection.

So,

what would

you say

about

7.3?

Okay.

So,

7.3

is

obviously

similar

to what

we

saw

in

chapter

4.

And

if I'm

not

mistaken,

even the

topology

is

exactly

the

same

with

that

picture

we

saw

previously.

So,

this is

the
initial
setup
which
acts
also
as a
review
about
how
these
different
components
of
POMDP
generative
models
needs to
be
described
in
such
situations.
But
ultimately,
the
specific
case
study
we
come across
in this
section
is
the
attempt
to model
the

behavior
of
a
mouse
and
a
rat
and
a
teammate.
So,
especially
teammates
containing
an
aversive
stimulus
in one
arm
and an
attractive
stimulus
on the
other.
So,
this
can
act
as
a
toy
example
to
use
this
kind
of
probabilistic

modeling
to
describe
these
situations.

Thanks.

So,
that leads
us right
to figure
7.4.

Here's a
visualization
of the
situation

with

the

rat

in this

case

where

there's

a

pleasant

and

aversive

stimuli

on each

end

of a

decision

point

and

there's

also

an

epistemic

opportunity

to
receive
some
information
about
the
context
that
the
animal
is
in.
And
so,
that
setting
is
described
for
both
the
case
with
white
on
the
left
and
black
on
the
right
and
black
on
the
left
and

those
are
shown
in
terms
of
their
differences
in
the
matrices
the
explicit
specification
of
the
generative
model.
Visualizations
show
some
of the
slices
of
the
B
variable
which
reflect
different
transition
probabilities.
C
represents
the
preferences
which
are

expressed
over
the
observable
states.

D
reflects
the
priors
on
the
different
states
that
need
priors.

7.4
what
would
you
say
about
this?

Okay
so
in
7.4
it
builds
up
on
the
previous
section
and
adds
other
elements

that
we
previously
saw
in
chapters
3
and
sorry
2
and
4
which
is
how
the
exact
formulation
for
expected
free
energy
can be
used
sorry
variation
free
energy
can be
used
to
formulate
the
trade-off
between
the
I mean
information

seeking
and
or at
least
between
the epistemic
value and
information
seeking.
so
here
it
uses
again
that
rat
example
in a
bit
more
more
extended
and
elaborate
form
to
formulate
the
epistemic
value
of
observing
Q
in a
given
location
and
figure

7.7

is a
representation
of
this
situation
but
another
situation
that's
been
let me
see
yeah
in

7.9

another
case study
discussed
here
is
the
situation
of
the
saccadic
eye
movements
and
because
it is
something
that can
be quite
successfully
described
or characterized
in terms

of
information
seeking
versus
the
epistemic
value
and the
situation
here
is
let me
see
yeah
shown
visually
in
figure
7.9
which
clearly
shows
how
our
visual
saccotic
eye
movements
can be
described
in such
a way
as to
kind of
trace
trajectory

of our

eye
movement
among
different
regions
of the
visual
space
so
and how
the information
we gather
from
a given
region
can
affect
the
subsequent
trajectories
of our
saccadic
eye
movements
so
yeah
that's
basically
the
main
premise
of
this
section
I guess
nice
great
7.5

what
would you
say about
it
okay
so
7.5
again
adds
another
dimension
to
the
previous
formulations
and
this
time
we
get
to
update
the
generative
models
by
learning
and
the
so
the
generative
models
for this
situation
is a bit
more
complicated

than the
previous
ones
because
it now
needs to
account
for
a
mechanism
or
a way
to
update
the
matrices
we had
before
so
in the
previous
situations
we
didn't
account
for
learning
per se
but
here
we
directly
update
our
general
sorry
the word
update

can be
confusing
here
we
get to
somehow
improve
our
generative
models
to
accommodate
for
these
updating
accounts
and
yeah
so
this
the situation
here
or the
case
study

here
which
somehow
elucidate
the way
that
the
learning
can be
accounted
for
with these

models
is
again a toy
example
of a
creature
in a
simple
world
black
and white
tiles
which
kind of
tries to
find
a
path
to
reach
a
given
destination
a
certain
destination
so
it is
more
complicated
than
the
situation
we had
for
the rat
example
because

it only
had
I mean
simple
trajectories
that needed
to
traverse
but
here
the
creature
or the
agent
in this
case
needs
to
do
lots
more
learning
and
information
seeking
and so
on
so
all the
previous
elements
is
kind of
combined
in this
example
and
it's a

really
good
example
to see
how the
different
components
of active
inference
can be
connected
to each
other
nice
and
7.6
hierarchical
or deep
inference
first
a box
7.3
interlude
on
structure
learning
boxed
off
topic
and a
lot
to say
but
structure
learning
broadly
refers
to

learning
the
structure
about
a
model
using
the
same
types
of
methods
that
you
might
to
do
inference
on
for
example
a
more
observable
sensor
data
reading
something
like
that
this
section
works
towards
the
idea
of
nested

inference
or
multi-scale
modeling
what would
you say
about
figure
712
okay
so
again
this
situation
is
I think
the most
complex
situations
of this
chapter
which
builds
up
from
the
previous
sections
and
this
time
it
adds
another
layer
to
accommodate
!

for
the
inferences
that
happen
in
different
time
steps
so
in
this
case
we
have
multi-time
or
multi-scale
inference
and learning
happening
both
at the
levels
of
learning
and
at the
levels
of
information
seeking
so
this
is
represented
in
figure

7.12

which

represents

how

kind

of

this

fractal

generative

model

can be

seen

as

a

component

in

this

multi-scale

bigger

generative

or

as a

kind

of

leaf

in

this

bigger

generative

model

so

it

can

be

seen

as

a

lower

level
inference
happening
at
the
leaf
level
going
up
to
the
hierarchy
and
influencing
sorry
collaborating
and the
whole
process
of
learning
and
inference
at
the
higher
level
so
yeah
I guess
that's
somehow
summarizes
this
figure
so
if you
have

anything
to add
that's
that's
great
it's
an
example
of
the
composability
of
generative
models
what
we've
talked
about
and had
Toby
Sinclair
Smith
describe
as
the
compositional
cognitive
cartography
and just
what
kinds of
connectors
can and
can't
you do
and how
can that
motif

that the
discrete
time
model
introduces
and then
the rest
of these
features
including
action
and learning
and so
on
get layered
in on
top
what can
you do
with that
7.13
gives another
example
do you want
to say
anything about
it or
maybe
continue
on
yeah
so the
case study
here is
the example
of
linguistic
language

learning
through
reading
so
not
language
learning
maybe
just
what
happens
in
reading
yeah
yeah
in
comprehension
so what
happens
when
reading
in an
anticipatory
way
the
words
that
comes
each
after
the
other
so
why
this
kind
of
situation

can
be
most
successfully
characterized
with
this
kind
of
modeling
because
it
involves
different
scales
of
learning
and
comprehension
both
at the
level
of
I mean
reading
at the
level
of
somehow
observing
the
letters
and then
going
on to
the words
and then
word groups

and so
on
so
yeah
that's
really
interesting
way to
again
combine
all of
those
elements
into a
single
unified
model
to see
how those
different
time scales
slow and
fast time
scales
operate
together
to
build
this
more
encompassing
model
generative
model
of
the
situation
more

encompassing
generative
model
of
the
situation
more
encompassing
generative
model
of
the
situation